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|  Republic of the Philippines<br>Department of Agriculture<br>BUREAU OF FISHERIES AND AQUATIC RESOURCES<br><b>REGIONAL FISHERIES LABORATORY XII</b><br>J. Catolico St., Lagao, General Santos City |                                                                       |                                 |                          |
| Document Type<br><b>Work Instruction</b>                                                                                                                                                                                                                                           | Revision No:<br>0                                                     | Date Adopted:<br>13 August 2019 | Page No:<br>Page 1 of 10 |
| Document Code:<br><b>WI-02-MIC-01</b>                                                                                                                                                                                                                                              | <b>ESTIMATION OF MEASUREMENT UNCERTAINTY IN MICROBIOLOGICAL TESTS</b> |                                 |                          |

## 1. Introduction

This document provides the procedure for the estimation of measurement uncertainty (MU) in microbiological testing which applies to quantitative methods used in the analyses. This procedure is taken from the National Public Health Service for Wales. Health Protection Agency. *Uncertainty of Measurement in Testing*. Issue No. 5. (2005).

## 2. Responsibilities

- 2.1 Technical Manager
- 2.2 Quality Assurance Manager
- 2.3 Laboratory Analysts

## 3. General Principles

- 3.1 No attempt is made in the reference to assess sample variation, although it is expected that the laboratory have an understanding of the distributions of organisms within matrices they test and to take this into account when subsampling.
- 3.2 In general, no measurement or test is perfect and the imperfections (such as pipettor variation) may give rise to error of measurement in the result. Thus, in an enumeration test the bacterial count obtained is only an approximation of the actual count and so the result is incomplete unless accompanied by a statement of the uncertainty associated with the count.
- 3.3 Uncertainty of measurement is a parameter associated with the result of a measurement that characterizes the dispersion of values that could reasonably be attributed to the measurand (such as a count/g).
- 3.4 As uncertainty of measurement is a parameter associated with the result of a measurement it is only applicable to enumeration methods. However, factors that contribute to the uncertainty of the result of presence/absence tests should be identified.
- 3.5 There are two types of evaluations that may be used to calculate uncertainty:
  - **Type A evaluations of uncertainty** are done by calculations from a series of repeated observations, using statistical methods.
  - **Type B evaluations of uncertainty** are derived from other sources e.g. from calibration data.
- 3.6 In a microbiological test, the factors contributing to a type B evaluation usually form part of a type A evaluation and so may not need to be considered separately. They also usually represent such a small contribution to the combined standard uncertainty that they do not make a significant contribution. For microbiological testing purposes, the combined standard uncertainty in a type A evaluation is not significantly different from

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| Prepared by:                                      | Reviewed by:                                           | Approved by:                               |
| CHARISSE ZIANNE N. LIBRES<br>Analyst Microbiology | MA. MAURICE A. BANQUILING<br>Quality Assurance Manager | EUGENE GAY B. JAMORA<br>Laboratory Manager |

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the standard uncertainty; the type B components can therefore be ignored in the calculation but identified and shown to be under control.

3.7 To simplify the evaluation of uncertainty it is necessary to exclude “less than” results and to use counts over 100/g because logs of such results are usually distributed in an approximately normal fashion.

#### 4. Specific Contributory Factors to Uncertainty in Microbiology

##### 4.1 Technical competence, bias and experience

- all stages of processing a sample, operation of equipment and qualitative or quantitative reading of tests
- variation between and within members of staff

##### 4.2 Sample

- homogeneity of original sample source
- sample sent to a laboratory
- test portion used in the subsample analysis
- precision and accuracy of balance or volumetric equipment
- non-uniform distribution of micro-organisms between subsamples or test portions
- time, transport and storage conditions between sampling and testing

##### 4.3 Homogenization of sample

- degree of heterogeneity of suspensions made from the sample
- clumping of micro-organisms
- uneven distribution of micro-organisms
- insufficient mixing

##### 4.4 Dilutions

- accuracy of pre- measured volumes or weights of dilution fluids
- volume of dilution fluid used
- degree of mixing at each dilution step
- number of steps in a serial dilution
- precision, accuracy and appropriate use of diluting equipment
- pipette volume used
- micro-organisms adhering to pipettes

##### 4.5 Media and reagents

- quality of media/reagents
- accurate weighing of media/reagent
- water quality including pH and conductivity

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- personal error in preparation and use of culture media (including appropriatetemperature when adding supplements)
- heat processing and control
- adequate mixing
- degree of dryness of solid media
- performance of media and reagents such as selectivity and sensitivity
- shelf life

#### 4.6 Inoculation of media

- volume of inoculum
- equipment used in dispensing, spreading and filtering
- temperature of molten agar in pour plate techniques

#### 4.7 Incubation conditions

- Duration
- Temperature
- Humidity

#### 4.8 Reading and interpretation of results

- recognition of target colonies
- number of colonies counted
- dilutions chosen for counting (one dilution or more than one dilution)
- proportion of colonies confirmed

### 5. Evaluation of Uncertainty of Measurement for Microbiological Testing

In the microbiological test methods, the sampling procedure to provide laboratory samples is not part of the process and the sample is “as received in the laboratory”. A subsampling step to provide the test portion is included in the test procedure. For the analysis of fish and fishery products only part of the sample submitted to the laboratory is examined, but for water tests, the entire laboratory sample is examined, in which case the distribution of the organisms within the matrix is less likely to affect the outcome of the result.

#### 5.1 The measurand for each quantitative test in the microbiological analysis of water, fish and fishery products are identified and listed below.

- Aerobic Plate Count and *S. aureus* Count– “calculation of the number of bacteria in fish and fishery products (in cfu/g) within the 25 g and 10<sup>-1</sup> dilution of the test portion which are capable of growing aerobically under the test conditions of an incubation period of 48 ± 2 hours at 35°C ± 1°C”

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- Yeasts and Molds Count - “calculation of the number of yeasts and molds in fish and fishery products (in cfu/g) within the 25 g and  $10^{-1}$  dilution of the test portion which are capable of growing aerobically under the test conditions of an incubation period of  $120 \pm 2$  hours at  $25^{\circ}\text{C}$ ”
- Heterotrophic Plate Count - “calculation of the number of heterotrophic bacteria (cfu/ml) in water or melted ice samples which are capable of growing under the test conditions of an incubation period of  $48 \pm 2$  hours at  $35^{\circ}\text{C} \pm 1^{\circ}\text{C}$ ”
- Total Coliform Count of Water - “calculation of the number of total coliforms (MPN/100 ml) in water or melted ice sample which are capable of growing under the test conditions of an incubation period of  $48 \pm 2$  hours at  $35^{\circ}\text{C} \pm 1^{\circ}\text{C}$ ”
- Fecal Coliform and *E. coli* Count of Water- “calculation of the number of fecal coliforms and *E. coli* (MPN/100 ml) in water or melted ice sample which are capable of growing under the test conditions of an incubation period of  $48 \pm 2$  hours at  $44.5^{\circ}\text{C} \pm 0.2^{\circ}\text{C}$ ”
- Total Coliform Count of Fish and Fishery Products - “calculation of the number of total coliforms (MPN/g) within the 25 g and  $10^{-1}$  dilution in fish and fishery products which are capable of growing under the test conditions of an incubation period of  $48 \pm 2$  hours at  $35^{\circ}\text{C} \pm 1^{\circ}\text{C}$ ”
- Fecal Coliform and *E. coli* Count of Fish and Fishery Products- “calculation of the number of fecal coliforms and *E. coli* (MPN/g) within the 25 g and  $10^{-1}$  dilution in fish and fishery products which are capable of growing under the test conditions of an incubation period of  $48 \pm 2$  hours at  $45.5^{\circ}\text{C} \pm 0.2^{\circ}\text{C}$ ”

5.2 For each test method, list the components that contribute to the uncertainty of the result and establish that they are under control in LF W04-03-01 (Lists of Contributory Factors to Uncertainty in Microbiological Tests).

5.3 For General Enumeration Methods:

- a) identify the individual components of uncertainty
- b) demonstrate that they are under control
- c) state that the individual components are inherent within the evaluation process
- d) identify the start point for uncertainty evaluations
- e) use reproducibility data as the basis for expression of uncertainty

5.4 Evaluation of Uncertainty Using Test Results

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The use of reproducibility or repeatability data in the evaluation of uncertainty in food microbiology is supported by the international consensus recognizing the difficulties in calculating the contributions of individual components.

- a) The laboratory uses real samples for provision of data but the full range of sample matrices examined by the laboratory will be covered. Data from other sources (actual samples, spiked samples, etc.) may also be used in the uncertainty assessment.
- b) Sample examination will be done during routine laboratory operation and this will reflect all the operating factors contributing towards uncertainty. The duplication will be performed by different operators/analysts (reproducibility). The replication will be performed using as many variables as possible, such as different operators/analysts, different pipettors, different incubators/water baths operating at the same temperature, different batches of media, etc. which will help reflect the variations that may occur naturally in the laboratory.
- c) Replicate (2 or more) results from each sample will be entered into an EXCEL spreadsheet and the data will be analyzed to obtain the reproducibility standard deviation  $RSD_R$  (see Annex A for details). Results to at least three significant figures will be used in order to minimize rounding errors. Levels of similar magnitude should be used, for example  $10^1 - 10^2$  for indicator parameters or  $10^3 - 10^5$  for aerobic colony counts.
- a) Further results will be added to the spreadsheet on a regular and rolling basis, so that the laboratory's ongoing performance will be assessed in a contemporaneous way. At least 30 sets of samples are required in order to apply a coverage factor of 2 to obtain the expanded uncertainty. "Less than" values (below the limit of detection) will not be included in the computation.
- b) The uncertainty is assessed annually to ensure that it still reflects the conditions in the laboratory such as staffing and equipment in use. If a significant change in the operating conditions of the laboratory, such as significant staff changes or changes of equipment, or if significant changes are made to the method used (such as a change of plating medium), the effect of these changes on UM should be evaluated.
- c) Raw data are listed in LF W04-03-02 (Estimation of Measurement Uncertainty in Microbiological Tests).
- d) Calculations of the MU are also filed in LF W04-03-02 (Estimation of Measurement Uncertainty in Microbiological Tests) and the soft copy of the Excel spreadsheet is locked and protected with a password.

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### 5.5 Presence/Absence Tests

Presence/absence tests do not result in an enumeration; therefore uncertainty of measurement cannot be calculated using the above approach. However the factors affecting the variability of the results is considered and shown to be under control. For each pathogen or organism of concern, the lower limit of detection that can be achieved consistently is determined and the sensitivity and specificity of the test is calculated.

### 5.6 Most Probable Number (MPN) Tests

If contributory components can be shown to be under control, the theoretical 95% confidence intervals given in MPN tables can be used to provide the expanded uncertainty range.

## 6. Expression of Results

6.1 When reporting the result of a test, the extent of information given depends mainly on the customer's requirements, the specification and the intended use of the result. The methods used to assess the results are available in the report or in the records of the test. Information includes sufficient documentation to demonstrate how the uncertainty was obtained and enable a repeat of the calculation if necessary.

6.2 When reporting enumeration tests, the result of the test is reported together with the expanded uncertainty appropriate to the 95% level of confidence, and expressed to no more than two significant figures. The results is accompanied by the following statement if at least 30 sets of data have been used and a normal distribution is assumed:

*The reported expanded uncertainty is based on a standard uncertainty multiplied by a coverage factor of  $k = 2$  (or other appropriate  $t$  value), providing a level of confidence of approximately 95%.*

6.3 For fewer sets of data and if the conditions of normality apply, a  $t$  distribution may be assumed. In this case the following statement is used:

*The reported expanded uncertainty is based on a standard uncertainty multiplied by a coverage factor  $k = XX$ , which for a  $t$  distribution with  $YY$  effective degrees of freedom provides a level of confidence of approximately 95%*

6.4 When reporting results of MPN tests, report the test result and the 95% confidence limits (obtained from tables).

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| CHARISSE ZIANNE N. LIBRES<br><b>Analyst Microbiology</b> | MA. MAURICE A. BANQUILING<br><b>Quality Assurance Manager</b> | EUGENE GAY B. JAMORA<br><b>Laboratory Manager</b> |

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6.5 For presence/absence tests, report the sensitivity and specificity or concordance together with the lowest level of organisms that can be recovered consistently.

## 7. Assessment of Compliance with Specification

7.1 If a statement of compliance is required, for example against a numerical limit specified in a standard, the count obtained in the test is expanded by the uncertainty interval at a level of confidence of 95% before comparison with the numerical limit.

7.2 For microbiological tests, maximum limits are usually specified. Compliance is achieved if the expanded count is lower than the maximum limit. A statement of compliance with the specification is made. If the maximum limit is exceeded even when the measured count is decreased by half the uncertainty interval, a statement of non-compliance with the specification is made. If the measured count exceeds the maximum limit but when decreased by half the uncertainty interval it is less than the maximum limit, it is not possible to confirm compliance or non-compliance with the specification. The test result and expanded uncertainty is reported together with a statement that compliance was not demonstrated. If the measured result is below the maximum limit but when expanded exceeds the maximum limit a similar statement is also made. For example:

*The measured result is above (below) the specification limit by a margin less than the measurement uncertainty; it is therefore not possible to state compliance based on the 95% level of confidence. However the result indicates that non-compliance (compliance) is more probable than compliance (non-compliance) with the specification limit.*

## 8. Forms

LF W04-02-02, Lists of Contributory Factors to Uncertainty in Microbiological Tests

LF W04-02-01, Estimation of Measurement Uncertainty in Microbiological Tests

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## ANNEX A: Sample Calculation of Relative Standard Deviations of Reproducibility ( $RSD_R$ ) to Assess the Uncertainty of Measurement for Counts

Annex A describes calculations of the relative standard deviation of reproducibility ( $RSD_R$ ) using a type A evaluation to estimate the uncertainty of measurement for counts.

### ➤ Duplicate Counts

If duplicate counts are made on the same test material by two different operators, then they can be used to provide an estimate of the  $RSD_R$ . While the calculations can be performed on the raw counts it is usual to take common logarithms of the count. Note that common logarithms are logarithms to the base 10 e.g.  $\log_{10}100 = 2.00$ . If we consider for example that for operators, A and B, counts of 1089 and 1211 were obtained respectively, when testing the same material. The  $RSD_R$  is calculated as follows:

- Take common logarithms of the counts  
 $x_A = \log_{10}(1089) = 3.03702787976$   
 and  $x_B = \log_{10}(1211) = 3.08314414314$
- The  $RSD_R$  is calculated using the following formula:

$$RSD_R = \sqrt{2 \left( \frac{x_A - x_B}{x_A + x_B} \right)^2}$$

- Substituting  $x_A$  and  $x_B$  into equation (1) we obtain:

$$\sqrt{2 \cdot \left( \frac{(3.037028 - 3.083144)}{(3.037028 + 3.083144)} \right)^2} = \sqrt{2 \cdot \left( \frac{-0.046116}{6.120172} \right)^2} = 0.01065627$$

- The  $RSD_R$  is multiplied by 100 to obtain a notional percentage; in this example the  $RSD_R$  is then expressed as 1.0656%.
- Compute for the combined reproducibility  $RSD_{CR}$  using the formula below:

$$RSD_{RC} = \sqrt{\frac{\sum_{i=1}^n RSD_{R_i}^2}{n}}$$

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### Application of the $RSD_{RC}$ in Uncertainty of Measurement (UM)

1. The  $RSD_{RC}$  as calculated above is the average relative standard deviation of reproducibility, after common logarithms ( $\log_{10}$ ) of the observed counts have been taken. To obtain the UM for a particular count,  $c$ , use the equation below:

$$UM = \log_{10}(c) \pm k * RSD_{RC}$$

where  $k$  is the appropriate coverage factor, usually equal to 2.

2. This will produce a range of the common logarithms of the count after UM has been applied. This range is that which could reasonably be expected if this material were to be tested by different operators within this laboratory.
3. To obtain the UM limits on the natural count scale, the antilog of these two  $\log_{10}$  values can be taken to obtain UM limits on the count scale. This can either be done using computer software or a hand-held calculator, the mathematical formula for the antilog is: **antilog( $v$ ) =  $10^v$**  e.g. if the resultant quantity was 5.034, then the antilog would be  $105.034 \approx 108143$
4. Example: If the  $RSD_{RC}$  is 0.011 and the count is  $6.76 \times 10^4$ , and we assume a coverage factor of  $k = 2$ . The common logarithm of  $6.76 \times 10^4$  is 4.8299. The upper and lower limits are:

$$4.8299 \pm 2 * 0.011 = 4.8299 \pm 0.022 = 4.8079 \text{ to } 4.8519$$

5. If a range for counts were required, it would be  $6.43 \times 10^4 - 7.11 \times 10^4$ . Note that this results in an asymmetric range around the count value.
6. **Important Points to Note:**
  - a) Use only actual counts, so exclude any results that are either "less than" or "greater than" values.
  - b) During any intermediate stages in the calculations, e.g. when transforming counts to  $\log_{10}$  values, calculating the mean, etc. try and keep figures as accurate as possible and only round the final results to the desired precision.
  - c) It is useful to include columns in the Excel spreadsheet that allow the date of testing, sample matrix and identity of the operators to be recorded.
  - d) Lock Excel formulas. Refer to SOP-13, Control of Records for locking of Excel Formulas.

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## 9. REFERENCE

National Public Health Service for Wales. Health Protection Agency. *Uncertainty of Measurement in Testing*. Issue No. 5. (2005)

## 10. DOCUMENTS

SOP-13, Control of Records

## 11. DOCUMENT HISTORY

| Revision No. | Issue Date | Original Entry | Revised as | Recall Date |
|--------------|------------|----------------|------------|-------------|
|              |            |                |            |             |
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| Prepared by:<br><br>CHARISSE ZIANNE N. LIBRES<br><b>Analyst Microbiology</b> | Reviewed by:<br><br>MA. MAURICE A. BANQUILING<br><b>Quality Assurance Manager</b> | Approved by:<br><br>EUGENE GAY B. JAMORA<br><b>Laboratory Manager</b> |
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